



SpiderVault 120.0

Model: Octane 120.0 · 120 kVA Three-Phase Solar Hybrid UPS

BATTERY INPUT	
Battery Type	Lithium Ion
Battery Capacity	120 kWh
Battery Minimum Voltage	624 V
Battery Maximum Voltage	874 V
Max. Charging / Discharging Current	240 A
Max. Charging / Discharging Power	125 kW
Charging Curve	3 Stages / Equalization

PV STRING INPUT	
Max. DC Input Power	156 kW
No. of MPPT Trackers	10
PV Input Voltage	800 V (160 – 1000 V)
MPPT Voltage Range	200 – 850 V
Start-up Voltage	160 V
PV Input Current	(36+36+36+36+36) × 2 A
Max. PV Isc	(55+55+55+55+55) × 2 A
No. of Strings per MPPT	(2+2+2+2+2) × 2

AC OUTPUT			
Rated AC Output & UPS Power	120 kW	Max. AC Output Power	132 kW
Peak Power (Off Grid)	1.5 × Rated Power for 10 s	Grid Type	Three Phase
AC Output Rated Current	174 A	Max. AC Current	191.6 A
Max. Continuous AC Passthrough	400 A	Current Harmonic Distortion	THD < 3%
Output Frequency & Voltage	50 / 60 Hz · 230 / 400 VAC · 220 / 380 VAC		

EFFICIENCY	
Maximum Efficiency	97.6%
MPPT Efficiency	> 99%

PROTECTION	
PV Input Lightning Protection	Yes
Anti-Islanding Protection	Yes
PV String Input Reverse Polarity Protection	Yes
Over Current Protection	Yes
Output Shorted Protection	Yes
Temperature Protection	Yes
Output Over Voltage Protection	DC Type II AC Type III
DC Switch	Yes

GENERAL DATA	
Mounting	Ground Mount
Topology	Transformer-less
IP Level	IP20
Working Temperature	Full power at or below 45°C
Cooling — Battery	Advanced Nano PCM + Forced Convection
Cooling — Inverter	Intelligent Air Cooling
Display	Colourful Touch LCD
AC Connector	OT Terminal
Communication Mode	Ethernet RS485 CAN

CERTIFICATIONS & STANDARDS
GRID REGULATION
• CEI 0-21, VDE-AR-N 4105, NRS 097
• IEC 61727, G99, G98
• VDE 0126-1-1, RD 1699, C10-11
SAFETY REGULATION
• IEC/EN 62109-1, IEC/EN 62109-2
EMC
• IEC/EN 61000-6-1 & 61000-6-2
• IEC/EN 61000-6-3 & 61000-6-4

120 kW
Rated Power

97.6%
Peak Efficiency

120 kWh
Battery Capacity

10
MPPT Trackers